PERFECT OBJECTS ON A QUASI-COMPACT OPEN OF AN AFFINE SCHEME BECOME RESTRICTIONS OF PERFECT OBJECTS AFTER ADDING A SHIFTED FINITE LOCALLY FREE SUMMAND

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Allowed standard prerequisites (editorial).

- A perfect complex is pseudo-coherent, and the pseudo-coherent lifting lemma produces a bounded above complex of finite free modules over the affine scheme restricting to the object. invoked at proof lines 3534-3539. Provides the lifting complex F^bullet with E isomorphic to its restriction.
- A perfect object on a quasi-compact open has finite tor amplitude [a, b], and the last kernel of a bounded above complex with flat restriction is flat by the cohomology lemmas. invoked at proof lines 3540-3548. Shows the kernel sheaf K|_U = F is flat, hence finite locally free.
- The stupid truncation triangle of a complex, the splitting criterion for a distinguished triangle whose connecting map vanishes, and the vanishing of Ext out of a perfect object into a bounded quasi-coherent complex in bounded degrees. invoked at proof lines 3558-3566. Produces the splitting P|_U = F[-r-2] plus E for r very negative.

Source reference. Original lemma 08CQ: sources/stacks-chow/tags.txt, source lines 10610–10610. Cited at proof lines 3536 and 3540 for pseudo-coherence and the tor amplitude of E; the lifting and kernel arguments are inside the counted span.

Source reference. Original lemma 08EE: sources/stacks-perfect/pinned-original-source.tex, source lines 3321–3321. Cited at proof lines 3537-3539 to lift E to F^bullet; the kernel and splitting arguments are inside the counted span.

Source reference. Original lemma 08CH: sources/stacks-chow/tags.txt, source lines 10602–10602. Cited at proof lines 3547-3548 for flatness of K|_U; the finite local freeness conclusion is inside the counted span.

Source reference. Original lemma 05P2: sources/stacks-chow/tags.txt, source lines 7326–7326. Cited at proof lines 3549-3550 for finite local freeness of F; the strictly perfect representative is inside the counted span.

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Source reference. Original lemma 05QT: sources/stacks-chow/tags.txt, source lines 7387–7387. Cited at proof lines 3563-3564 to split the truncation triangle; the vanishing of the connecting map is inside the counted span.

Source reference. Original lemma 09M4: sources/stacks-perfect/pinned-original-source.tex, source lines 3465–3465. Cited at proof lines 3565-3566 for the vanishing of the connecting map for r very negative; the splitting conclusion is inside the counted span.

Lemma 0.2. *Let X be an affine scheme. Let $U \subset X$ be a quasi-compact open. For every perfect object E of $D(\mathcal{O}_U)$ there exists an integer r and a finite locally free sheaf $\mathcal{F}$ on U such that $\mathcal{F}[-r] \oplus E$ is the restriction of a perfect object of $D(\mathcal{O}_X)$.*

Proof. Say $X = \mathrm{Spec}(A)$. Recall that a perfect complex is pseudo-coherent, see Cohomology, Lemma 08CQ. By Lemma 08EE we can find a bounded above complex $\mathcal{F}^\bullet$ of finite free A-modules such that E is isomorphic to $\mathcal{F}^\bullet|_U$ in $D(\mathcal{O}_U)$. By Cohomology, Lemma 08CQ and since U is quasi-compact, we see that E has finite tor dimension, say E has tor amplitude in $[a, b]$. Pick $r < a$ and set

$$\mathcal{K} = \mathrm{Ker}(\mathcal{F}^r \to \mathcal{F}^{r+1}) = \mathrm{Im}(\mathcal{F}^{r-1} \to \mathcal{F}^r).$$

Since E has tor amplitude in $[a, b]$ we see that $\mathcal{F} = \mathcal{K}|_U$ is flat (Cohomology, Lemma 08CH). Hence $\mathcal{F}$ is flat and of finite presentation, thus finite locally free (Properties, Lemma 05P2). It follows that

$$\mathcal{F} \to \mathcal{F}^r|_U \to \mathcal{F}^{r+1}|_U \to \ldots$$

is a strictly perfect complex on U representing E. On the other hand, the complex $P = (\mathcal{F}^r \to \mathcal{F}^{r+1} \to \ldots)$ is a perfect complex on X. Using stupid truncations we obtain a distinguished triangle

$$P|_U \to E \to \mathcal{F}[-r-1] \to (P|_U)[1]$$

If the map $E \to \mathcal{F}[-r-1]$ is zero in $D(\mathcal{O}_U)$, then $P|_U = \mathcal{F}[-r-2] \oplus E$, see Derived Categories, Lemma 05QT. This will be true for $r \ll 0$ for example by Lemma 09M4. $\square$